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To cite this article: Takahiro Tamura and Takaaki Suzuki 2019 *Jpn. J. Appl. Phys.* **58** SDDL10

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Seamless fabrication technique for micro to millimeter structures by combining 3D printing and photolithography

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Received November 30, 2018; revised January 29, 2019; accepted February 6, 2019; published online June 10, 2019

This paper proposes a seamless fabrication technique for micro-electro-mechanical systems (MEMS) devices having structures from micrometer to millimeter with a single material by combining 3D printing and photolithography. The proposed fabrication technique combines the fabrication characteristics of 3D printing and photolithography. We fabricated a mold of a microfluidic device having micro- and millimeter structures and evaluated three characteristics of the proposed technique. From the evaluations, the micro- to millimeter structures were seamlessly fabricated with an error within $\pm 5\%$. The rough surface of the resist transferred from a 3D-printed master mold causes irregular reflection of UV light and reduction in patterning accuracy. The relation of object shape and fabrication of millimeter structures was evaluated. In addition, various advantages of the proposed fabrication technique were confirmed by demonstrating a fluid transportation test for the fabricated microfluidic device.

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1. Introduction

In the past decades, many micro-electro-mechanical systems (MEMS) devices have been developed^{1–7)} because of their many advantages such as miniaturization, power saving, higher functionality, and lower cost. These MEMS devices having structures from sub-micrometer to sub-millimeter are often fabricated by photolithography,^{3–7)} electron-beam lithography,^{2,6,8)} and embossing.^{9,10)} Recently, in addition to these fabrication techniques, 3D printing is beginning to be widely used^{11–16)} in the MEMS field to conveniently fabricate prototype devices and complex devices having structures from sub-millimeter to millimeter. However, in most cases, these fabrication techniques exist independently and each of them has the limitation of the fabrication size and shape. These limitations in fabrication often cause problems when fabricating devices. For example, in the field of bio-MEMS, connections between macroscale external devices and micro-scale fluidic devices (known as the macro-to-micro interface, interconnect, connector, or world-to-chip interface) have been reported as a problem.^{17–19)} Also, the fabrication of a micro-fluidic device having micrometer structures by 3D printing has been reported as a problem.^{20,21)}

These fabrication techniques are mapped in three aspects²²⁾ as shown in Fig. 1. The horizontal axis is relative accuracy, which is object size divided by tolerance. The right side of the figure indicates that accurate fabrication is performed on a design value. The vertical axis is object size and indicates the range of fabricable sizes using each fabrication technique. The red color indicates that arbitrary complex structures are fabricated. As shown in the figure, photolithography fabricates simple structures from sub-micrometer to sub-millimeter. However, it is difficult to fabricate 3D millimeter structures by photolithography²³⁾ due to several factors, e.g. film thickness distribution of deposited thick film, optical attenuation of photoresist in the vertical direction, diffraction of light, and depth of focus for thick film. On the other hand, consumer-grade desktop 3D printers fabricate structures from sub-millimeter to centimeter with high complexity.^{12,14)} However, it is difficult to fabricate micrometer structures precisely.^{20,21)} These two

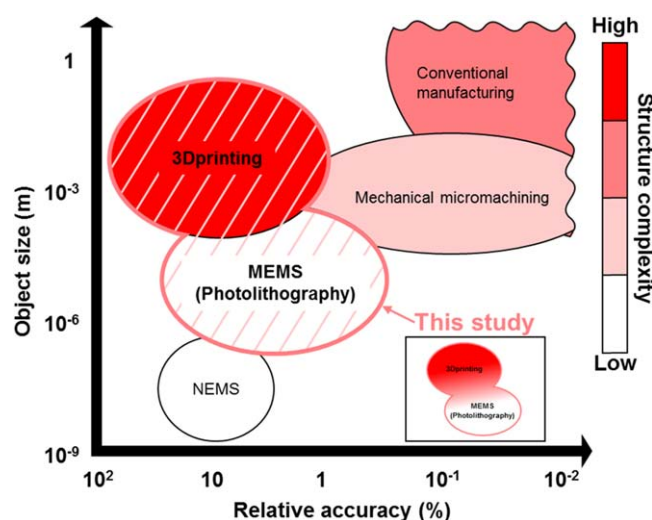


Fig. 1. (Color online) Purpose of this study. The horizontal axis is relative accuracy, the vertical axis is object size, and the color shows the ability to fabricate complex structures. Photolithography fabricates simple structures from sub-micrometer to sub-millimeter. On the other hand, consumer-grade desktop 3D printers fabricate sub-millimeter to centimeter structures with high complexity. The proposed fabrication technique realizes seamless fabrication from micrometer to millimeter with a single material by combining 3D printing and photolithography, indicated by hatching.

fabrication techniques overlap in the area of sub-millimeter but exist independently. Therefore, it is difficult for each technique to fabricate structures seamlessly from micrometer to millimeter.

In this study, we propose a seamless fabrication technique from micrometer to millimeter with a single material by combining 3D printing and photolithography, as indicated by the hatching in Fig. 1. 3D printing and photolithography, which are the base of the proposed technique, have different characteristics. The proposed fabrication technique combines these two techniques and has mixed characteristics of each technique. So, we fabricated a mold for a microfluidic device having micrometer and millimeter structures by the proposed technique and evaluated three characteristics: the fabrication accuracy, surface roughness, and object shape. In the

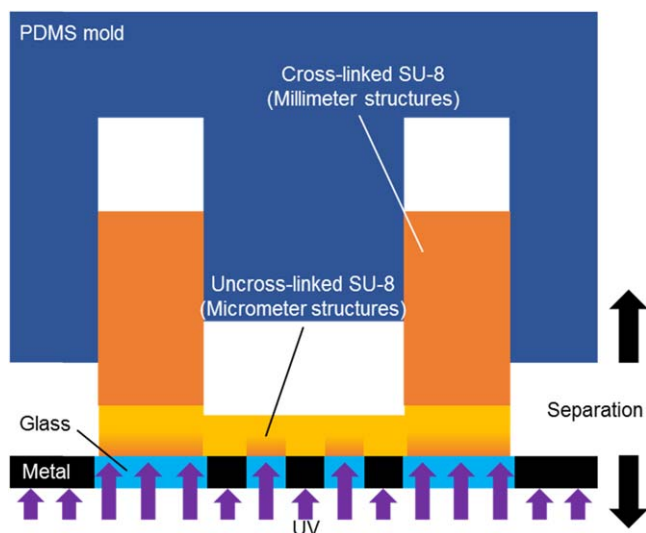


Fig. 2. (Color online) Principle of the proposed technique. SU-8 millimeter structures are fabricated by SU-8 molding with a PDMS mold that is replica-molded with a 3D-printed master mold. Micrometer structures are fabricated in the molded SU-8 by photolithography.

evaluation of the fabrication accuracy, the designed and measured values of the fabricated micrometer and millimeter structures are compared. Moreover, a microfluidic device was fabricated as one of several applications of the proposed technique to confirm the various advantages. Concretely, the proposed fabrication technique realizes seamless fabrication of micrometer and millimeter structures without assembly for a complex microfluidic device. In addition, the proposed technique realizes complex and fine structures by combining 3D printing and photolithography.

2. Experimental methods

2.1. Principle of fabrication technique

The proposed fabrication technique is based on a combination of 3D printing and photolithography using thick negative photoresist SU-8 (MicroChem Corp.). As shown in the schematic in Fig. 2, SU-8 millimeter structures are fabricated by SU-8 molding with a polydimethylsiloxane (PDMS) mold that is replica-molded with a 3D-printed master mold. SU-8 is widely used for fabricating MEMS devices as a thick negative photoresist.²⁴⁾ In addition, SU-8 micrometer structures are fabricated by photolithography. In other words, millimeter structures are fabricated by SU-8 molding with the 3D-printed master mold and micrometer structures are incorporated in the molded SU-8 by photolithography. So, a seamless fabrication technique of structures from micro- to millimeter made of a single material is realized by combining 3D printing and photolithography.

2.2. Fabrication process

Figure 3 shows a process flow of the proposed fabrication technique. Firstly, a master mold is designed with STL file format by 3D-CAD software (Inventor professional, Autodesk) and printed using a 3D printer (Form 2, Formlabs) with 25 μm height of one layer [Fig. 3(a)]. In the 3D printing, standard clear resin (FLGPCL04, Formlabs) is used. This master mold has structures from sub-millimeter to millimeter size precisely fabricated by a consumer-grade desktop 3D printer,^{20,21)} then a PDMS (SILPOT 184, Dow Corning) mold is fabricated by replica molding^{25,26)} [Fig. 3(b)]. Next, liquid SU-8 (SU-8 3005,

MicroChem) is poured into the millimeter structures and baked at 95 °C for 2 h [Fig. 3(c)]. After baking the SU-8, the level of SU-8 liquid surfaces decreases by losing the solvent.²⁷⁾ Therefore, pouring and baking SU-8 is repeated until the distance between the PDMS mold surface and the baked SU-8 surface becomes 1 mm or less. After SU-8 cooling, ultraviolet (UV) light is exposed to the baked SU-8 with $\geq 2000 \text{ mJ cm}^{-2}$ (wavelength 365 nm) which is a sufficient exposure dose for the millimeter-thick SU-8 [Fig. 3(d)]. After that, SU-8 is baked at 65 °C for 1 min and 95 °C for 6 min. As a result, SU-8 millimeter structures are fabricated. Next, SU-8 (SU-8 3050, MicroChem) is coated with a thickness of 200 μm or more on a Cr-patterned glass substrate and baked at 95 °C for 2 h [Figs. 3(e) and 3(f)]. After SU-8 cooling, the SU-8-coated substrate and the PDMS mold are bonded. After that, pressure is applied to the substrate under a vacuum to degas²⁸⁾ the air inside the substrate and the PDMS mold for 10 min [Fig. 3(g)]. After degassing, pressure is applied with 0.1 kPa to the substrate and the SU-8 is baked at 95 °C for an hour using an oven [Fig. 3(h)]. As a result, the uncrosslinked SU-8 exceeds the glass transition temperature²⁷⁾ and flows into the patterns on the PDMS mold.²⁹⁾ After SU-8 cooling, the PDMS mold and the substrate are separated. After that, UV light is exposed with a normal exposure dose for an SU-8 sub-millimeter thickness from the back side of the Cr-patterned glass substrate³⁰⁾ [Fig. 3(i)]. The SU-8 pattern molded with the PDMS mold is designed to cover the Cr pattern on the substrate. Therefore, the shape of the micrometer structures is determined by photolithography. Then, the SU-8 is baked at 65 °C for 1 min and 95 °C for 6 min [Fig. 3(j)]. After cooling the SU-8 for 30 min, it is developed with SU-8 developer (MicroChem) for 10 min and then rinsed with isopropyl alcohol (Kanto Chemical) for 1 min, two times [Fig. 3(k)]. Through the process, micrometer structures and millimeter structures are seamlessly fabricated with the single material.

2.3. Evaluation

In general, 3D printing and photolithography, which are the base of the proposed technique, have different characteristics in terms of their object size, surface roughness, and object shape, as shown in Table I. The proposed fabrication technique combining 3D printing and photolithography has mixed characteristics from each technique.

By using the proposed fabrication technique, a mold of a microfluidic device having micro- and millimeter structures was fabricated as shown in Fig. 4(a). Also, the characteristics of the proposed fabrication technique (1)–(3) were evaluated as shown in Fig. 4(b). First, to evaluate the fabrication accuracy, the structures on the fabricated device were observed using a scanning electron microscope (SEM) (JCM-5700LV, JEOL) and designed and measured values of the structures were compared. Second, to evaluate the effect of the resist surface roughness on the micrometer structures, two devices fabricated using a surface-treated master mold and a master mold without surface treatment were compared. Third, the relationship between object shape and fabrication was evaluated by three types of structures with three aspect ratios. Finally, a PDMS microfluidic device was fabricated by soft lithography²⁵⁾ using the microfluidic mold fabricated by the proposed technique. The PDMS microfluidic device and a glass substrate were bonded by oxygen-plasma treatment³¹⁾ using a dry etching system

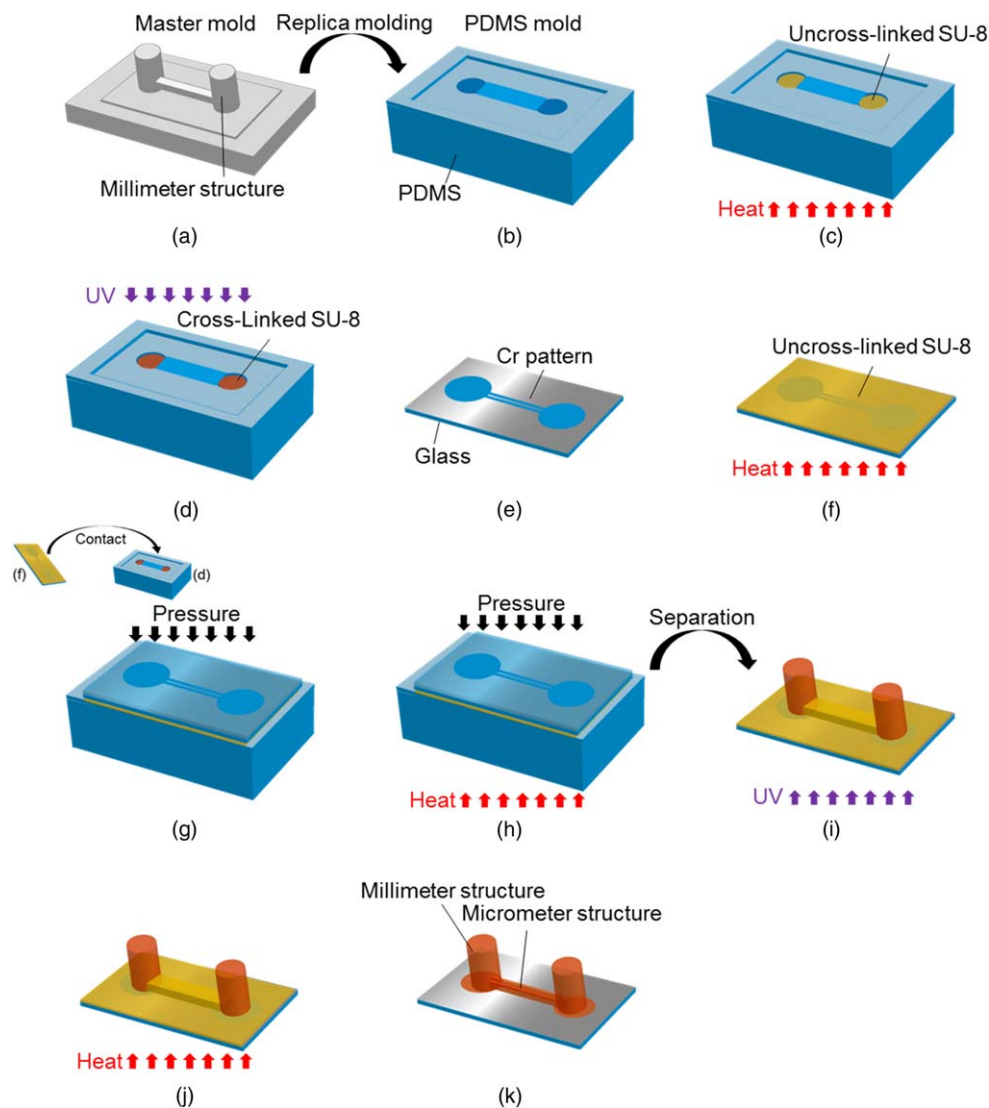


Fig. 3. (Color online) Process flow of the proposed technique. (a) Master mold fabricated with a 3D printer. (b) PDMS mold fabricated by replica molding. (c) Liquid SU-8 is poured and baked. (d) Soft-baked SU-8 exposed to UV light. (e), (f) Coated and soft-baked SU-8 prepared on a Cr-patterned glass substrate. (g) Pressure and vacuum are applied. (h) Bonded SU-8 is baked under pressure. (i) PDMS mold and substrate are separated, then soft-baked SU-8 is exposed to UV light from the back side. (j) Post-exposure baking of SU-8. (k) Development of SU-8.

Table I. Characteristics of fabrication technique. AR: aspect ratio.		
	3D printing	Photolithography
(1) Object size (fabricate with accuracy)	Sub-mm to cm	Sub- μm to sub-mm
(2) Surface roughness	Micrometer-order	Nanometer-order
(3) Object shape	Complex, high AR	Simple, low AR

(ES401, Nippon Scientific). A suspension having microspheres with a diameter of $2\text{ }\mu\text{m}$ (G0200, Thermo Fisher) was injected in the channel of the fabricated microfluidic device using a syringe pump (KDS270, KD Scientific) at the flow rate of $1\text{ }\mu\text{l min}^{-1}$ and the channel was observed using an inverted system microscope (IX-71, Olympus) and a charge-coupled device camera (DP72, Olympus).

3. Results and discussion

3.1. Fabrication accuracy

To evaluate the fabrication accuracy, the structures on the fabricated device were observed using SEM, and designed and measured values of the structures were compared. The

prototype mold has a channel with micrometer line-and-spaces and millimeter interfaces. Figure 5(a) shows a photograph of the prototype mold for the microfluidic device, and Figs. 5(b)–5(d) and 5(e) show SEM images of the line-and-spaces and interfaces of the fabricated device. As shown in Fig. 5, the micrometer structures and the millimeter structures were seamlessly and simultaneously fabricated with the single material. In addition, Table II shows the comparison of the measured values that were obtained from the SEM images and the designed values. As shown in Table II, both micrometer and millimeter structures were fabricated with an error within $\pm 5\%$. Photolithography and 3D printing have a different range of fabrication sizes, and in the proposed technique, micrometer structures are fabricated by photolithography and millimeter structures are fabricated by 3D printing. As a result, high fabrication accuracy is obtained with a wide range.

3.2. Surface roughness

The effect of the resist surface roughness on micrometer structures was evaluated. As we mentioned, a master mold was fabricated by 3D printing. As a result, the surface of the

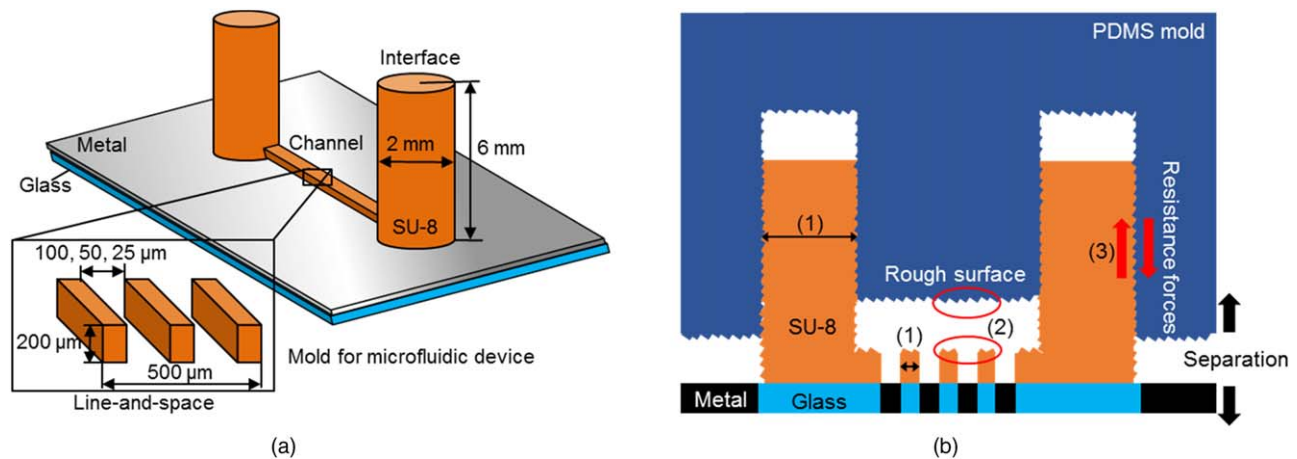


Fig. 4. (Color online) Evaluation of fabrication characteristics. (a) Schematic of a prototype microfluidic device mold. (b) Fabrication characteristics of the proposed fabrication technique. (1) Object size, (2) surface roughness, (3) object shape.

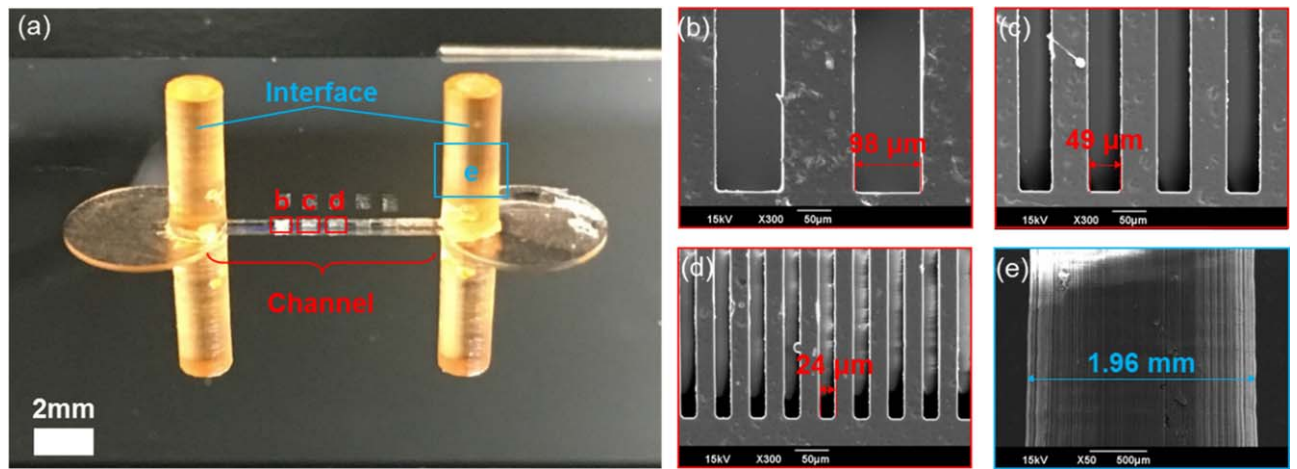


Fig. 5. (Color online) Evaluation of fabrication accuracy. (a) Photograph of the prototype mold for a microfluidic device. (b)–(d) SEM images of micrometer line-and-space in the channel from the top surface. Line widths of designed values are (b) 100 μm, (c) 50 μm, (d) 25 μm. (e) SEM image of cross-sectional view of millimeter interface. Designed value of interface diameter is 2 mm.

Table II. Comparison between designed and measured dimensions.

	Line-and-space in the channel (μm)			Interface diameter (mm)
Designed size	100	50	25	2.00
Measured size	98	49	24	1.96
Error	−2%	−2%	−4%	−2%

master mold is rough^{20,32)} and this roughness is transferred to the SU-8 surface. In this evaluation, two devices fabricated using a polished mold and an unpolished mold were compared to evaluate the effect of surface roughness on micrometer structures. The mold surface was polished with a cloth using liquid metal polish (Pikal, Nihon Maryo-Kogyo) until the surface had luster. Figure 6 shows SEM images of the fabricated devices. The devices shown in Figs. 6(a) and 6(b) were fabricated with the polished master mold and unpolished master mold, respectively. The top image shows the SEM image from above of the device, the bottom left image shows the SEM image from diagonally above with high magnification, and the bottom right figure shows the behavior of the UV light when the micrometer structures are exposed. From Fig. 6(a), the surface of the fabricated device using the polished mold is relatively smooth, and the

micrometer structures were fabricated without roughness. On the other hand, the surface of the device fabricated with the unpolished mold is rough, and the fabricated micrometer structures are also rough, as shown in Fig. 6(b). This is because irregular reflection occurred on the rough resist surface. UV light is incident from the back side, then passes through the resist and reaches the resist surface. When the resist surface is rough, UV light irregularly reflects at the rough resist surface and exposes masked areas. As a result, rough micrometer structures were fabricated as shown in Fig. 6(b). From the evaluation, it is necessary to perform surface treatment on the master mold to fabricate micrometer structures precisely.

3.3. Object shape

The object shape of the fabricated millimeter structures was evaluated. As shown in Fig. 3, millimeter structures are fabricated by molding with a 3D-printed master mold. As a result, millimeter structures could break at the separation of a PDMS mold and a substrate [Figs. 3(h) and 3(i)]. In this evaluation, three types of structures were fabricated with three aspect ratios to evaluate millimeter structure fabrication. Figure 7(a) shows the cross-sectional views of the fabricated devices arranged by the shape of millimeter structures and the aspect ratio. Additionally, Fig. 7(b) shows the equipment

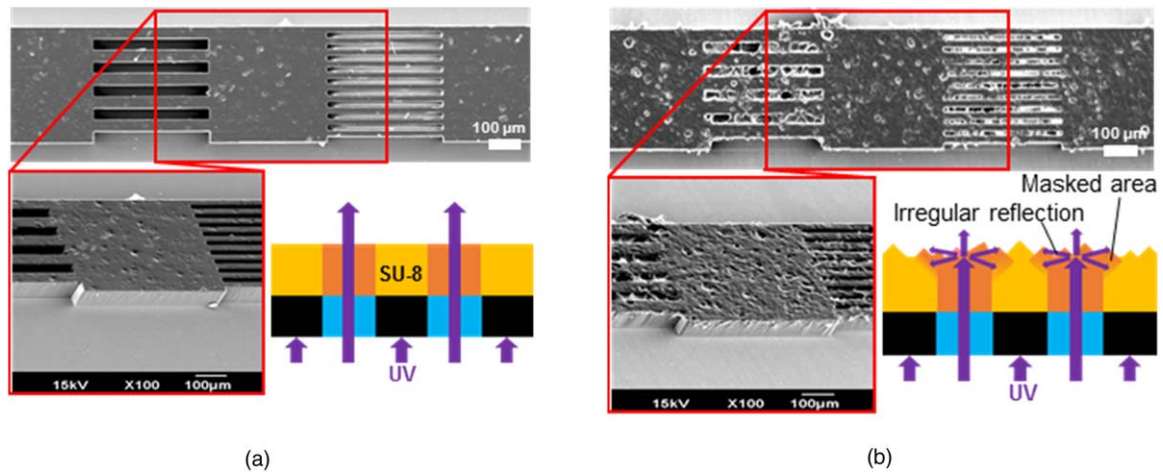


Fig. 6. (Color online) Effect of resist surface roughness on micrometer structures. (a) SEM images of the fabricated device using a polished mold. (b) SEM images of the fabricated device using an unpolished mold. Irregular reflection occurred at the rough resist surface and the masked area is also exposed.

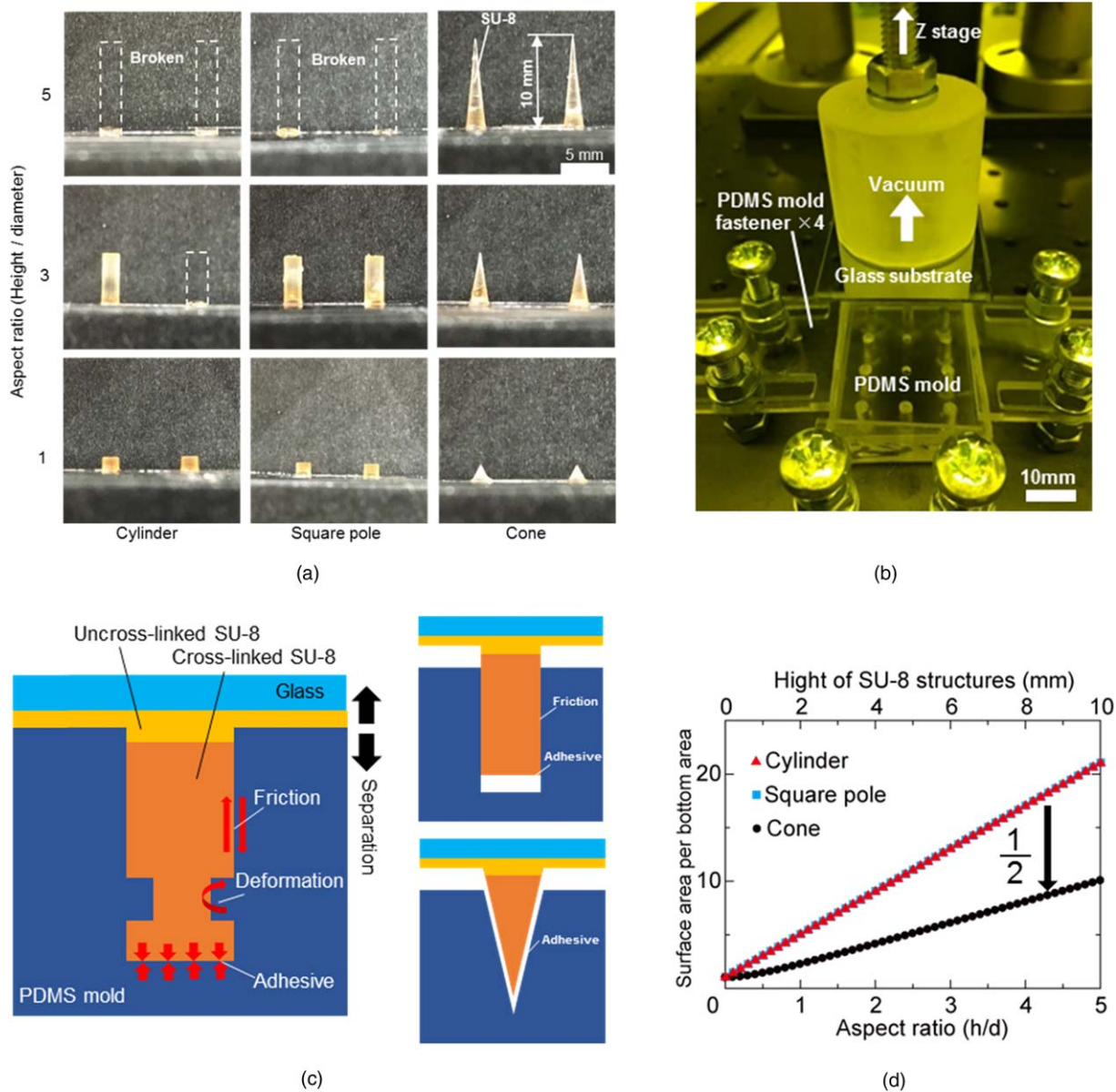


Fig. 7. (Color online) Object shape and fabrication of structures. (a) Cross-sectional views of the fabricated structures. Three types of millimeter structures (cylinder, square pole, and cone) were fabricated with three aspect ratios (AR: 1, 3 and 5) to evaluate millimeter structure fabrication. (b) Equipment for separation of PDMS mold and a glass substrate. (c) Schematic of the three types of resistance forces acting on millimeter structures when separating a PDMS mold and a substrate. (d) Surface area per bottom area loading the resistance forces as a function of the aspect ratio of the millimeter structures.

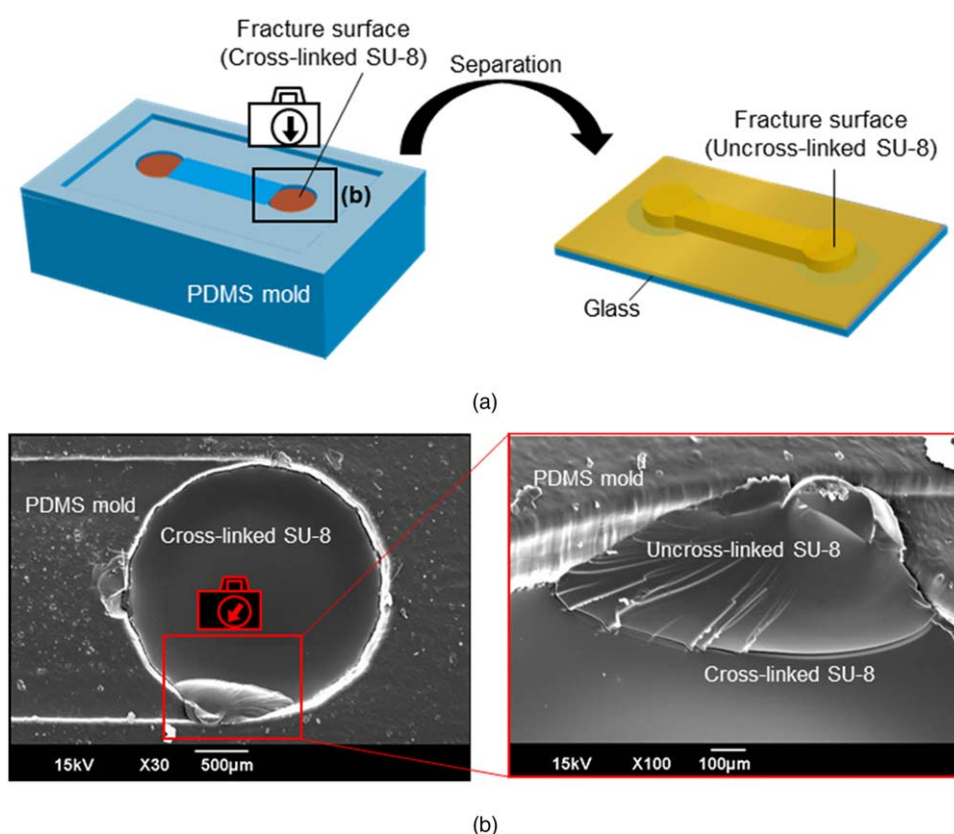


Fig. 8. (Color online) Observation of fracture surface of millimeter structure. (a) Schematic of the SEM observation point of the millimeter structures remaining in the PDMS mold after being broken on separation, and the crosslinked SU-8 fracture surface of the structure. (b) SEM images of the crosslinked SU-8 fracture surface. The left image was observed from above and the right image was observed from diagonally above with high magnification.

used for the separation of a PDMS mold and a glass substrate with SU-8 millimeter structures. The PDMS mold is fixed on the stage using four fasteners and a glass substrate is fixed by negative pressure, then the glass substrate is vertically pulled up by the Z stage.

As shown in Fig. 7(a), regarding the aspect ratio 5 with 10 mm height structures, only the cone structure was fabricated and the other two structures were broken when separating the PDMS mold and the substrate. There are two reasons for this. The first is resistance forces acting on the millimeter structures upon separation. As shown in the left of Fig. 7(c), there are a maximum of three types of resistance forces (adhesive force, friction force, and force required to deform a PDMS mold) acting on millimeter structures when separating a PDMS mold and a substrate. From the difference in the shape, the adhesive force and friction force act on the cylinder and square pole [upper right of Fig. 7(c)], whereas only the adhesive force acts on the cone [bottom right of Fig. 7(c)]. The second reason is the difference of surface area where resistance forces act. Figure 7(d) shows the surface area of the millimeter structures. The horizontal axis is the aspect ratio of the millimeter structures, the vertical axis is the surface area per bottom area. The plots show that the red triangle is the cylinder, the blue square is the square pole, and the black circle is the cone. As shown in the figure, the surface area per bottom area of the cone where resistance forces act is half that of the other structures. For these reasons, the SU-8 cone structure could be fabricated with aspect ratio 5, 10 mm height.

In addition, the fracture surface of the millimeter structure which remained in the PDMS mold was observed using SEM as shown in Fig. 8(a) and SEM images were obtained as shown in Fig. 8(b). The left image of Fig. 8(b) was observed from above and the right image was observed from diagonally above with high magnification. From the SEM images the breaking of the millimeter structure basically occurred at the boundary between the crosslinked SU-8 at the mold side and the uncrosslinked SU-8 at the bonding step as shown in Fig. 3(g). Since the millimeter structures are fabricated by pouring SU-8 into the PDMS mold and molding, the end face of the molded SU-8 has a smooth meniscus shape as shown in Fig. 8(b). Therefore, the crosslinked SU-8 and the uncrosslinked SU-8 are the bonding of smooth faces and it couldn't withstand a peeling force of high aspect ratio millimeter structures in the process.

From the evaluation, for fabricating millimeter structures, we should design easy-to-fabricate structures like the cone, and modify the shape or properties of the boundary between the crosslinked SU-8 and the uncrosslinked SU-8 at the bonding step.

3.4. Application

A microfluidic device was fabricated as an application and evaluated by a fluid transportation test. Figure 9 shows the fluid transportation test; the left image shows the injection of microsphere suspension into the fabricated microfluidic device and the right image shows the fluorescent microscopic image of the channel during the fluid transportation. As shown in the right image, flowing microspheres were

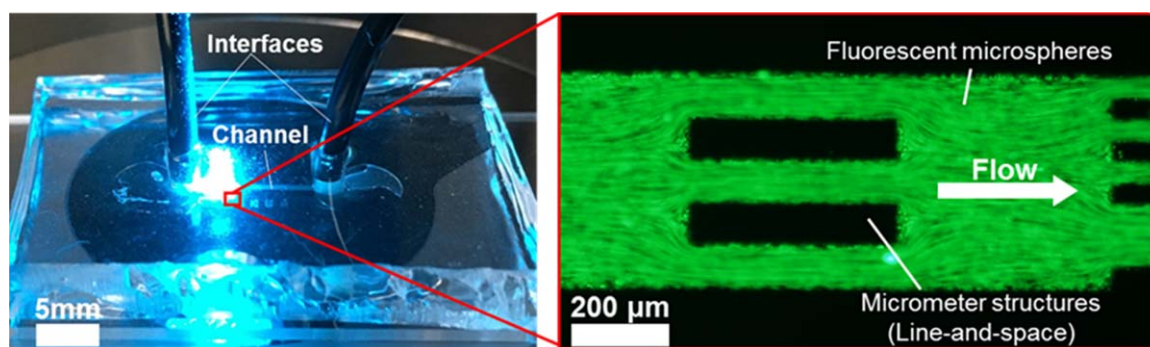


Fig. 9. (Color online) Fluid transportation test. Left image is a photograph of the PDMS microfluidic device fabricated by the proposed technique. The microfluidic device has a channel of width $500\ \mu\text{m}$ and height $200\ \mu\text{m}$. The channel has line-and-space patterns of $100\ \mu\text{m}$, $50\ \mu\text{m}$, and $25\ \mu\text{m}$. Right image is the fluorescent image of fluid transportation test. The suspension of microspheres with a diameter of $2\ \mu\text{m}$ was injected into the microfluidic device using a syringe pump at the flow rate of $1\ \mu\text{l min}^{-1}$.

observed in the channel having the micrometer line-and-spaces pattern.

Since the millimeter interfaces and the micrometer channel were seamlessly fabricated by our fabrication technique, it is not necessary to fabricate the interfaces after soft lithography. In addition, the dead volume is reduced by seamlessly fabricating the interfaces and the microchannel. Moreover, the fabricated device worked efficiently as a microfluidic device without suspension leakage.

The cross-sectional shape of the fabricated microchannel is a rectangle that can be fabricated by general photolithography. Additionally, the cross-section of a microchannel can be determined by 3D printing. So, it will be possible to fabricate micrometer structures in a millimeter fluidic channel having a more complex cross-sectional shape.

4. Conclusions

In this study, we fabricated a mold of a microfluidic device having micrometer structures and millimeter structures, and evaluated three characteristics of the proposed fabrication technique. First, micrometer to millimeter structures were seamlessly fabricated with an error within $\pm 5\%$ in the evaluation of fabrication accuracy. Second, the effect of resist surface roughness on micrometer structures was seen in the evaluation of surface roughness. Third, in the evaluation of object shape, a $10\ \text{mm}$ high SU-8 structure, difficult for general photolithography, was fabricated. Finally, the advantages of the proposed fabrication technique were confirmed by the fluid transportation test.

Acknowledgments

This research was partially supported by JSPS Science Research Grant (JP17H03196) and JST PRESTO (Grant No. JPMJPR15R3).

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